USSOCOM Technical Experimentation Event

Safety Guide

The purpose of this guide is to ensure safety risks associated with the operation of equipment to be demonstrated at a USSOCOM technical experimentation event are identified. Following a rigorous safety process will ensure proper mitigations are identified and implemented - facilitating a safe demonstration.

The desired format for submitting a record of safety risks is a Deliberate Risk Assessment (DRA) worksheet. An example is DD Form 2977 which can be found at this link: DD2977. However, the https://sam.gov/ posting should be consulted. for specific requirements.

Firms developing material for the Department of War are encouraged to integrate system safety into the technology development process. Any outputs of this integration will be useful to the team managing the USSOCOM technical experimentation event. Examples of common safety analyses include Safety Assessment Reports, System / Subsystem Hazard Analyses, Laser Safety Evaluations, Interim/Final Hazard Classifications, Energetic Material Sensitivity Analyses, Fuse Safety Analyses, and Developmental Test Reports & User Documentation with Warnings/Cautions incorporated.

The typical reference for System Safety within USSOCOM & the Department of War is Military-Standard 882, "Standard Practice: System Safety". The current revision (MIL-STD-882E) is available publicly on the web.

The methodology described in MIL-STD-882E, to include the definitions of probability, severity & risk level, are utilized across all DOD Components & most defense contractors. Consider using these common terms and measures when developing hazard analyses. Submission of safety artifacts beyond the required DD2977 form is highly encouraged, as are efforts to tailor mitigations to the specific technology being demonstrated & the testing scenario.

For reference, the next page lists several areas of frequent concern with the general safety data requirements associated with each area.

If the system contains a laser, the following information should be readily available to the range Laser Systems Safety Officer:

Nominal Ocular Hazard Distance (NOHD) with/without magnification (As applicable)

Optical Density (OD) requirements
Design wavelength
Beam divergence
Output power
Any records of a verification of laser output (independent measurement preferred)

If the system contains energetic materials (e.g. propellants, pyrotechnics, explosives), the following information should be made available to the experimentation team:
Record of an Interim Hazard Classification (IHC)
Any evaluation of Energetic materials [if novel] (e.g. sensitivity test results)
Any evaluation of Fuzing / Safe & Arm / Ignition Safety systems
Recommended standoff distances, surface danger zones and/or range fans
Summary of any prior live fire testing

If the system contains a Radio-Frequency / Magneto-Inductive (or other) emitter, the following information should be made available to the experimentation team:
Number of frequencies requested and band
Transmitter output power Antenna gain
Any “Radiation Hazard” (RADHAZ) analyses performed
Any established Safe Stand-off Distances (SSDs) for Ordnance, Personnel & fueling operations